

Introductory Real Analysis: Square Roots - Exam Problems

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1 Exam Problems

1.1 Exam Beta

1.1.1 Problem 1.

Evaluate the following: (3 points)

$$\sqrt{\frac{36}{25} \cdot \frac{81}{144} \cdot 0,36}$$

First, let $0,36 = \frac{36}{100}$, then split the square root into a product of fractions

$$= \sqrt{\frac{36}{25}} \cdot \sqrt{\frac{81}{144}} \cdot \sqrt{\frac{36}{100}} = \frac{6}{5} \cdot \frac{9}{12} \cdot \frac{6}{10} = \frac{6 \cdot 9 \cdot 6}{5 \cdot 12 \cdot 10} = \frac{324}{600} = \frac{27}{50} = \boxed{0,54}.$$

$$\begin{aligned} & \sqrt{72 \cdot 32} \\ & \sqrt{2^8 \cdot 3^2} = 2^4 \cdot 3 = 16 \cdot 3 = 48. \end{aligned}$$

Generally,

$$\sqrt{a}$$

Evaluate the prime factorization under the root

$$a = p_1^{\alpha_1} \cdot p_2^{\alpha_2} \dots p_i^{\alpha_i}$$

Where p_i prime. Then all prime factors with even exponents $2\omega_j = \alpha_j$ come out of the square root with half its' exponents, ω_j . All prime factors with odd exponents come out as half of its' exponents minus 1, so for factors with odd exponents $2\beta_k + 1 = \alpha_k$, $p_k^{\beta_k}$ comes out of the root, p_k stays in. Therefore our final result is

$$\sqrt{a} = p_1^{\frac{\alpha_1}{2}} \cdot p_2^{\frac{\alpha_2}{2}} \dots p_m^{\frac{\alpha_m}{2}} \cdot q_1^{\frac{\phi_1-1}{2}} \cdot q_2^{\frac{\phi_2-1}{2}} \dots q_l^{\frac{\phi_l-1}{2}} \cdot \sqrt{q_1 \cdot q_2 \dots q_l}$$

With p_i primes with even exponents and q_l primes with odd exponents.

$$\begin{aligned} & \sqrt{90 \cdot 64} \\ & \sqrt{90 \cdot \frac{64}{10}} \\ & \sqrt{10 \cdot 9 \cdot \frac{64}{10}} \\ & \sqrt{9 \cdot 64} \\ & \sqrt{9} \cdot \sqrt{64} = 3 \cdot 8 = 24. \end{aligned}$$

1.1.2 Problem 2.

Evaluate the following expressions (2 points)

$$2\sqrt{72} - \sqrt{50} - 2\sqrt{8}$$

$$\sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}$$

$$\sqrt{50} = \sqrt{25 \cdot 2} = 5\sqrt{2}$$

$$\sqrt{8} = \sqrt{4 \cdot 2} = 2\sqrt{2}$$

$$2(6\sqrt{2}) - 5\sqrt{2} - 2(2\sqrt{2}) = 12\sqrt{2} - 5\sqrt{2} - 4\sqrt{2} = 3\sqrt{2}$$

$$\sqrt{12} - \sqrt{27} + \frac{1}{2}\sqrt{48}$$

We notice that all of the numbers are divisible by 3.

$$2\sqrt{3} - 3\sqrt{3} + \frac{1}{2}4\sqrt{3}$$

$$2\sqrt{3} - 3\sqrt{3} + 2\sqrt{3}$$

$$\sqrt{3}.$$

1.1.3 Problem 3.

Simplify the following expressions. (2 points)

$$\sqrt{30} \cdot \sqrt{20}$$

$$\sqrt{30 \cdot 20} = \sqrt{600}$$

$$600 = 2^3 \cdot 3 \cdot 5^2$$

$$= \sqrt{2^2 \cdot 5^2 \cdot 2 \cdot 3} = 2 \cdot 5 \cdot \sqrt{6} = 10\sqrt{6}$$

$$\sqrt{63} \cdot \sqrt{28}$$

$$\sqrt{1764} = ?.$$

$$1764 = 2^2 \cdot 3^2 \cdot 7^2$$

$$\sqrt{2^2 \cdot 3^2 \cdot 7^2} = 2 \cdot 3 \cdot 7 = 42.$$

1.1.4 Problem 4.

Evaluate the expression. (3 points)

$$(2\sqrt{18} + \sqrt{54}) \cdot (3\sqrt{6} - \sqrt{27} - \sqrt{3})$$

Simplify every radical

$$2\sqrt{18} = 2 \cdot 3\sqrt{2} = 6\sqrt{2}$$

$$\sqrt{54} = 3\sqrt{6}$$

$$\sqrt{27} = 3\sqrt{3}$$

Then the expression becomes

$$(6\sqrt{2} + 3\sqrt{6}) \cdot (3\sqrt{6} - 3\sqrt{3} - \sqrt{3})$$

$$(6\sqrt{2} + 3\sqrt{6}) \cdot (3\sqrt{6} - 4\sqrt{3})$$

After simply algebraically expanding the brackets like with any expression, we get

$$54 + 36\sqrt{3} - 36\sqrt{2} - 24\sqrt{6}.$$

1.1.5 Problem 5.

Calculate the exact value of the following expression. (3 points)

$$(\sqrt{2 + 2\sqrt{3}} - \sqrt{-2 + 2\sqrt{3}})^2$$

We notice that our expression is of the form $(A - B)^2$. Thus expanding yields one of form $A^2 - 2AB + B^2$

$$\underbrace{(2 + 2\sqrt{3})}_{A^2} - \underbrace{2\sqrt{(2 + 2\sqrt{3})(-2 + 2\sqrt{3})}}_{2AB} + \underbrace{(-2 + 2\sqrt{3})}_{B^2}$$

The outer terms combine immediately:

$$4\sqrt{3} - 2\sqrt{(2 + 2\sqrt{3})(-2 + 2\sqrt{3})}$$

Now we simplify the product under the root. We notice that it is a difference of squares $(a + b)(a - b) = a^2 - b^2$ with $a = 2\sqrt{3}$, $b = 2$.

$$(2\sqrt{3} + 2)(2\sqrt{3} - 2) = 12 - 4 = 8.$$

$$4\sqrt{3} - 2\sqrt{8} = 4\sqrt{3} - 4\sqrt{3} - 4\sqrt{2} = \boxed{4(\sqrt{3} - \sqrt{2})}.$$

1.1.6 Problem 6.

Calculate the exact value of the following expression. (3 points)

$$\sqrt{48 + 6\sqrt{15}} + \sqrt{48 - 6\sqrt{15}}$$

We will omit the details here, ask any LLM. The final results, for checking purposes, is $6\sqrt{5}$.

1.1.7 Problem 7.

Rationalize the denominator.

$$\frac{2}{\sqrt{5} - 2}$$

The conjugate of $\sqrt{5} - 2$ is $\sqrt{5} + 2$, therefore we will multiply the entire fraction by that.

$$\frac{2(\sqrt{5} + 2)}{(\sqrt{5} - 2)(\sqrt{5} + 2)}$$

$$2\sqrt{5} + 4.$$

$$\frac{28}{1 + \sqrt{5}}$$

Multiply by the conjugate

$$\frac{28(1 - \sqrt{5})}{(1 + \sqrt{5})(1 - \sqrt{5})}$$

We omit the small details. The final result is

$$7(\sqrt{5} - 1).$$

1.1.8 Problem 8.

Simplify the following expression. (3 points)

$$\sqrt{a + 2 + 4\sqrt{a - 2}}$$

1.2 Exam Alpha**1.2.1 Problem 1.**

Calculate the value of the following expressions (? points)

$$\sqrt{\frac{64}{225} \cdot \frac{100}{9} \cdot 2,25} = \frac{8}{15} \cdot \frac{10}{3} \cdot \frac{15}{10} = \frac{8}{3}$$

$$\sqrt{75 \cdot 48} = \sqrt{3 \cdot 25 \cdot 3 \cdot 16} = 3 \cdot 5 \cdot 4 = 60$$

$$\sqrt{2,5 \cdot 14,4} = \sqrt{\frac{25}{10} \cdot \frac{144}{10}} = \frac{5 \cdot 12}{10} = 6$$

1.2.2 Problem 2.

Evaluate the following expressions. (? points)

$$\sqrt{80} + \frac{1}{2} \cdot \sqrt{20} + 3\sqrt{45}$$

$$4\sqrt{5} + \frac{1}{2}2\sqrt{5} + 3 \cdot 3 \cdot \sqrt{5}$$

$$4\sqrt{5} + \sqrt{5} + 9\sqrt{5}$$

$$14\sqrt{5}$$

Evaluate the following expressions

$$5\sqrt{3} - \frac{1}{3} \cdot \sqrt{27} - \sqrt{48}$$

$$5\sqrt{3} + \frac{1}{3} \cdot 3 \cdot \sqrt{3} - 4\sqrt{3}$$

$$2\sqrt{3}.$$

1.2.3 Problem 3.

1 point

$$\sqrt{50} \cdot \sqrt{125} = 5 \cdot \sqrt{2} \cdot 5\sqrt{5} = 25\sqrt{10}$$

$$\sqrt{48}\sqrt{27} = 4\sqrt{3} \cdot 3 \cdot \sqrt{3} = 36$$

1.2.4 Problem 4.

$$(3\sqrt{3}-\sqrt{5})(\sqrt{20}+2\sqrt{12}+2\sqrt{3}) = (3\sqrt{3}-\sqrt{5})(2\cdot\sqrt{5}+2\cdot2\cdot\sqrt{3}+2\sqrt{3}) = (3\sqrt{3}-\sqrt{5})(6\sqrt{3}+2\sqrt{5}) = 2\cdot(3\sqrt{3}-\sqrt{5})$$

1.2.5 Problem 5.

$$(\sqrt{6-\sqrt{11}}+\sqrt{6+\sqrt{11}})^2 = 6-\sqrt{11}+6+\sqrt{11}+2\sqrt{(6-\sqrt{11})(6+\sqrt{11})} = 12+2\sqrt{36-11} = 12+10 = 22.$$

1.2.6 Problem 6.

$$\sqrt{17 - 4\sqrt{15}} + \sqrt{17 + 4\sqrt{15}}$$

Then 17 is $a^2 + b^2$ and $4\sqrt{15}$ is $2ab$, therefore $ab = 2\sqrt{15}$ Now we can make a table

a	2	$2\sqrt{3}$	$2\sqrt{5}$	1b	$\sqrt{15}$	$\sqrt{5}$	$\sqrt{3}$	$2\sqrt{15}$	$a^2 + b^2$	19	17	23	61
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Now we can continue our solution
 $2\sqrt{3} + \sqrt{5} + 2\sqrt{3} - \sqrt{5} = 4\sqrt{3}$

1.2.7 Problem 7.

$$\frac{2}{\sqrt{3} - 2} \cdot \frac{\sqrt{3} + 2}{\sqrt{3} + 2} = \frac{2\sqrt{3} + 4}{3 - 4} = -2\sqrt{3} - 4$$

$$\frac{12}{\sqrt{17} + 5} \cdot \frac{5 - \sqrt{17}}{5 - \sqrt{17}} = \frac{12(5 - \sqrt{17})}{25 - 17} =$$

1.2.8 Problem 8.

$$\sqrt{3 - \sqrt{5}}$$

We notice that

$$\frac{6 - 2\sqrt{5}}{2} = \frac{(\sqrt{5} - 1)^2}{2}$$

Now we can continue our proof

$$\sqrt{\frac{(\sqrt{5} - 1)^2}{2}} = \frac{\sqrt{5} - 1}{\sqrt{2}} = \frac{\sqrt{10} - \sqrt{2}}{2}$$